MTH 107(7) Dr. James Ashe
Final Exam Friday 5$^{\text{th}}$ December **Name:**

Unless stated otherwise, all problems are worth 4 points, and answers should be stated as decimals rounded to three decimal places. You must show your work to receive credit.

1. Using the symbolic representations
m: I love math.
h: It is hot in hades.
g: I went to the game.
express the following statements in symbolic form.

a. I went to the game or it is hot in hades.

b. If I love math then it is not hot in hades.

2. Consider the statement, "If Theano is a woman then Theano is mortal." Use (a), (b), and (c) to complete parts 2a and 2b.
(a) If Theano is not a woman then Theano is not mortal.
(b) If Theano is mortal then Theano is a woman.
(c) If Theano is not mortal then Theano is not a woman.

a. Identify the converse of the statement. _____

b. Identify the contrapositive of the statement. _____

3. For parts 3a and 3b, identify the negation of the provided statement.

a. Some students like football.
(a) All students like football.
(b) No students like football.
(c) Some students like football.
(d) Some students do not like football.

b. All pilots can fly a plane.
(a) All pilots can fly a plane.
(b) No pilots can fly a plane.
(c) Some pilots cannot fly a plane.
(d) Some pilots can fly a plane..

4. [8 points each] Use a truth table to determine if the following arguments are valid.

a. 1. $q \vee r$
2. r
$\therefore \sim q$

		$\boxed{1}$	$\boxed{2}$		$\boxed{c}$	$\boxed{\text{Argument}}$
q	r	$q \vee r$	r	$1 \wedge 2$	$\sim q$	$(1 \wedge 2) \to c$
T	T					
T	F					
F	T					
F	F					

b. r: You register early.
d: You receive a discount.

1. If you register early, then you will receive a discount.
2. You register early.

Therefore, you will receive a discount.

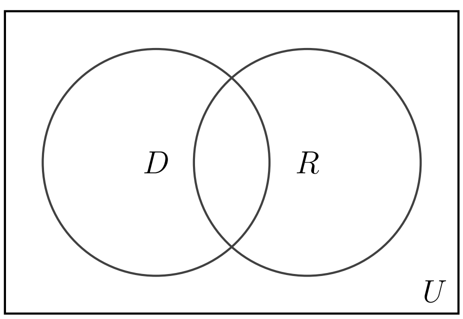
r	d	$\boxed{1}$	$\boxed{2}$	$1 \wedge 2$	$\boxed{c}$	$\boxed{\text{Argument}}$ $(1 \wedge 2) \to c$
T	T					
T	F					
F	T					
F	F					

5. Let $U = \{i, l, o, v, e, m, a, t, h\}$ be the universal set, and $A = \{l, o, v, e\}$ and $B = \{v, e, m, a, th\}$ be subsets of U. Find the indicated set, and draw a Venn diagram corresponding to the indicated set.

a. $A \cup B$ **b.** $A \cap B$ **c.** A'

6. A town has 1000 residents. Of the residents, 600 have voted Democrat, 500 have voted Republican, and 200 have voted for both parties.

a. Use the Venn diagram, and write the appropriate numbers in each region.



b. How many residents have voted Democrat but not Republican?

c. How many residents have voted Democrat or Republican?

d. How many residents have voted neither Democrat nor Republican?

7. [2 pts each] Find the value of each of the following:

a. $6!$

b. $\dfrac{10000!}{9999!}$

c. ${}_6P_4$

d. ${}_6C_4$

8. A parking decal consists of two letters (A-Z) followed by three numbers (0-9). If each letter and numbers can be repeated, how many different decals can be made using this method?

9. There are 10 pigs entered in the "That's some pig" contest. How many different ways can the judges award 1^{st}, 2^{nd}, and 3^{rd} place prizes? *Just write the formula.*

10. A three-person committee must be chosen from a group of three men and four women. How many committees are possible if it must consist of the following? *Just write the formula.*

a. three women.

b. one man and two women.

11. Two coins are tossed. Find each of the following:

a. the sample space

b. [2 points] the event, E_1, that at least one is tails.

c. [2 points] the probability of E_1.

12. A card is dealt from a well-shuffled, standard deck of 52 cards. Find the probability of being dealt each of the following:

a. an ace of hearts

b. an ace or a heart

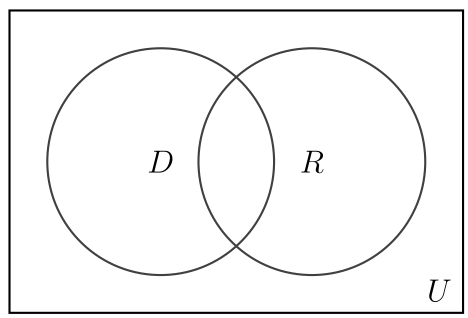
13. A bag of candy has 10 jelly beans. 6 of which are red and 4 of which are green. If you reach into the bag and randomly select 3 jelly beans, find the probability of the following: *Just write the formulas.*

a. all 3 are green.

b. exactly 2 are green.

14. Of all the residents in a town 60% have voted Democrat, 50% have voted Republican, and 20% have voted for both parties.

a. What is the probability that a randomly selected resident has voted Democrat but not Republican?



b. What is the probability that a randomly selected resident has voted Democrat or Republican?

15. A group of men and women were asked whether they preferred MacGyver or Oprah. Use the results provided in the table to answer the following questions.

	MacGyver	**Oprah**	**NA**
Men	40%	8%	2%
Women	15%	30%	5%

a. Find the probability that a man prefers Oprah.

b. Find the probability that a person who prefers MacGyver is a women.

16. In Prof. Phails history class, 40% of the students pass and the rest fail. Of the students who fail 55% are male and the rest are female. Of the students who pass 35% of the students are male and the rest are female.

a. draw a tree diagram describing the probability

b. find the probability that a student who fails is female.

c. find the probability that a student fails and is female.

17. Use the provided raw data set to complete the following:

1 2 2 2 3 4 2 0

a. Find the median.

b. Find the mean.

c. Find the mode.

d. Find the standard deviation.

18. Create a histogram using the provided frequency distribution table.

hours	**freq.**
$0 \leq x < 5$	5
$5 \leq x < 10$	10
$10 \leq x < 15$	15
$15 \leq x < 20$	10
$20 \leq x < 25$	2

19. Find the following probabilities,

a. $p(z < -.81)$ **b.** $p(z > -.81)$ **c.** $p(-1.15 < z < -.81)$

20. A population is normally distributed with mean 95 and standard deviation 10. Find the following probabilities.

a. $p(x < 80)$ **b.** $p(x > 107)$

21. Find c such that each of the following is true.

a. $p(z < c) = .6103$ **b.** $p(z > c) = .6103$ **c.** $p(z < c) = .43$

22. An IQ test is normally distributed with mean 100 and standard deviation 15. Find the percentage of the with a score less than 80.